



ACCELERATE DEEP LEARNING INFERENCE USING INTEL TECHNOLOGIES

OPTIMIZATION: TOOLS AND TECHNIQUES

March 2019

Core and Visual Computing Group

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Smart Video Workshop Overview

Introduction

1. Introduction to Intel technologies for deep learning inference
2. Hardware acceleration techniques

Each module contains a hands-on lab exercise that introduces various Intel technologies to accelerate computer vision application with hardware heterogeneity.

Intel®
Distribution of
OpenVINO™ 101

Hardware Acceleration

2. Basic End-to-End Object Detection Example

3./4./5. Hardware Acceleration with CPU, Integrated GPU, Intel® Movidius™ NCS, FPGA

Optimization

6. Optimization Tools and Techniques

Application

7. Advanced Video Analytics

Optimizing Computer Vision Applications

- Use OpenVINO:
 - Model Optimizer:
 - Use the right data type(FP16 or 32) for your target hardware and accuracy needs
 - Inference Engine:
 - Performance difference between using the inference engine (IE) and a non-optimized framework can be bigger than differences between accelerators
 - Set batch size
 - Use Async API while capturing video at the same time as doing the inference
- Other Recommendations:
 - Use/train a model with the right performance/accuracy tradeoffs. Performance differences between models can be bigger than any optimization you can do at the inference app level
 - Don't infer every frame if not needed
 - Use Intel® VTune™ Analyzer



Use the Correct Data Type

Use the Correct Data Type

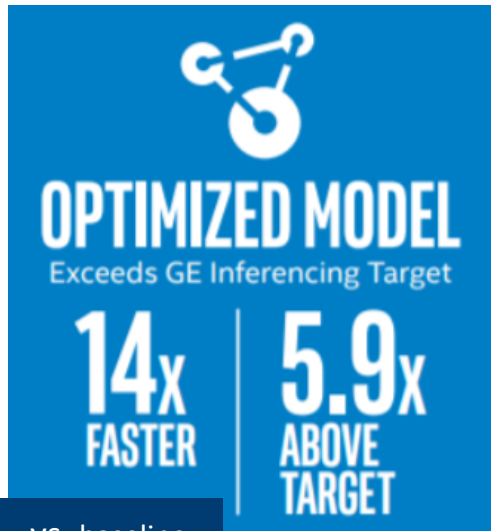
	FP32	FP16
CPU	yes	no
GPU	yes	recommended
Intel® Movidius™ Myriad™	no	yes
FPGA/DLA	no	yes

FPGA/DLA also supports FP11.



Use Inference Engine

Use an Optimized Inference Engine vs. a non-optimized framework on Intel® CPU



The Deep Learning Deployment Toolkit from Intel helps deliver optimized inferencing on Intel® architecture, helping bring the power of AI to clinical diagnostic scanning and other healthcare workflows

https://ai.intel.com/wp-content/uploads/sites/53/2018/03/IntelSWDevTools_OptimizeDLforHealthcare.pdf

The performance difference between running inference engine vs. unoptimized can be bigger than the difference between accelerators.

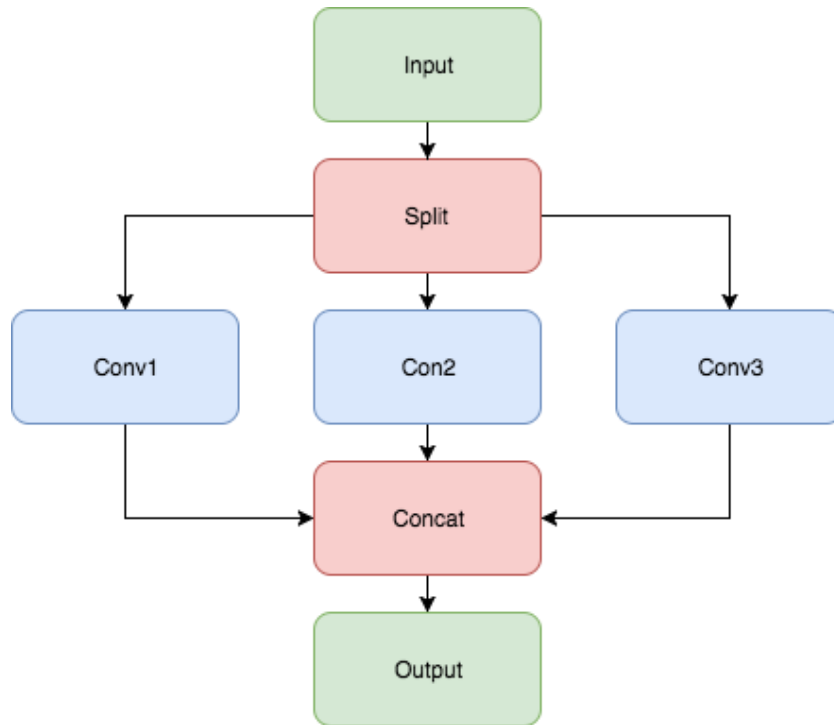
Why?

- The CPU plugin was developed in order to provide opportunity for high performance scoring of neural networks on CPU, using the Intel® Math Kernel Library for Deep Neural Networks (Intel® MKL-DNN).
- The CPU plugin uses OpenMP* in order to parallelize calculations.
- The CPU Plugin supports inference on Intel® Xeon® with Intel® AVX2 and AVX512, Intel® Core™ Processors with Intel® AVX2, Intel Atom® Processors with Intel® SSE.

Internal CPU Plugin Optimizations

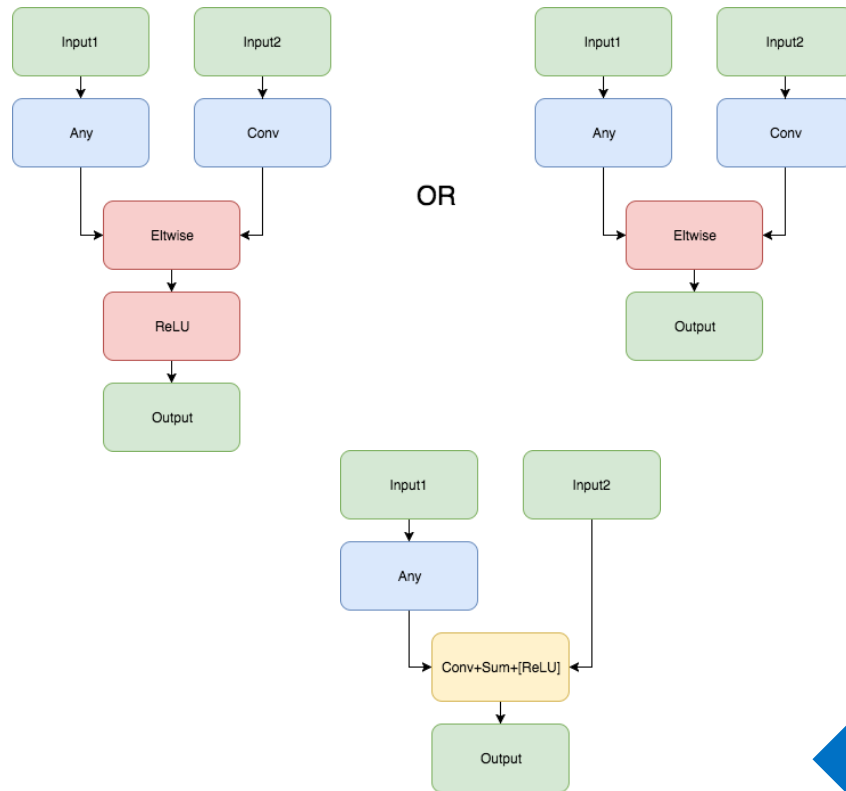
Merging of group convolutions.

- It means that if a topology contains the following pipeline →
- CPU plugin will merge it into one Convolution with the group parameter (Convolutions should have the same parameters).



Internal CPU Plugin Optimizations

- **Fusing Convolution with ReLU or ELU.** CPU plugin is fusing all Convolution with ReLU or ELU layers if these layers are located after the Convolution layer.
- **Fusing Convolution + Sum or Convolution + Sum + ReLU.** To improve performance, the CPU plugin fuses the following structure:



Set Batch Size

Set Batch Size

- To increase throughput, a convolutional neural network (CNN) could process the input data in batches.
- What is batch size?
 - If we have a 3-channel(RGB) images that are 256x256 pixels. A single image can be represented as a $3 \times 256 \times 256$ matrix. If we set the batch size to 10, then we are concatenating those 10 images together into a $10 \times 3 \times 256 \times 256$ matrix.
- What is the best Batch size? – It depends on the model and available compute power
 - Increase batch size could help to get a higher throughput which means better inference FPS, but if it goes too large, your device will run out of memory to hold it.



Use Async

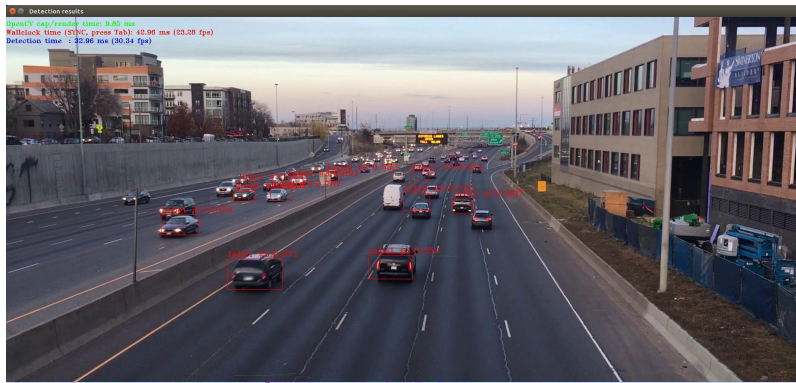
Inference Engine async API

- Improves overall frame-rate of the application.
- Executes a request asynchronously (in the background) and waits until ready, when the result is actually needed.
- Continues doing things on the host, while the accelerator is busy.
- The demo keeps two parallel infer requests, and, while the current is processed, the input frame for the next is captured. This essentially hides the latency of capturing, so the overall framerate is determined by the `MAXIMUM(detection time, input capturing time)` and not the `SUM(detection time, input capturing time)`.

Running Object Detection Sample SSD async

```
$ ./object_detection_demo_ssd_async -i /home/intel/workshopApr25/workshop-tutorials/test_content/video/cars_1920x1080.h264 -m /home/intel/workshopApr25/workshop-tutorials/test_content/IR/SSD/SSD_GoogleNet_v2_fp32.xml
```

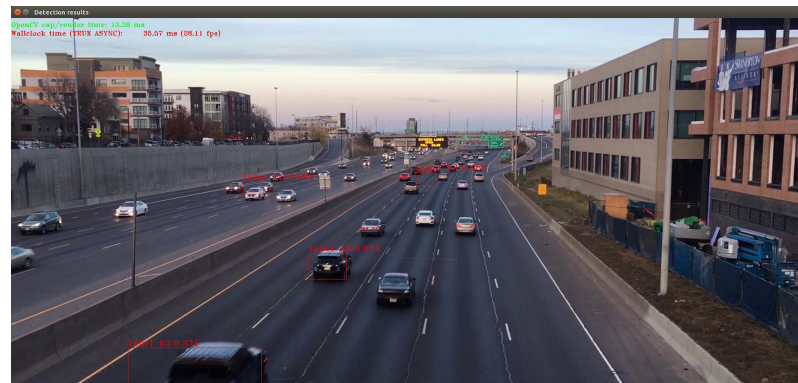
Synchronous Mode



Detection results

OpenCV cap/render time: 9.85 ms
Wallclock time (SYNC, press Tab): 42.96 ms (23.28 fps)
Detection time : 32.96 ms (30.34 fps)

Press **Tab** to Enable Asynchronous Mode



Detection results

OpenCV cap/render time: 13.26 ms
Wallclock time (TRUE ASYNC): 35.57 ms (28.11 fps)

Pick the Right Model

Use/Train a Model with the Right Performance plus Accuracy Tradeoffs.

Performance is based on many factors:

- Topography complexity/layer implementation plus scheduling
- Number of color channels (that is, BGR vs. grayscale)
- Model resolution

Exercise: Range of Model Performance

Focus on the Inference Timing

Complete this table in the hands-on exercise:

	CPU ms/frame	GPU ms/frame	Intel® Movidius™ Myriad™ X ms/frame
ssd512			
ssd300			
Mobilnet-ssd*			



Don't Infer If Not Needed

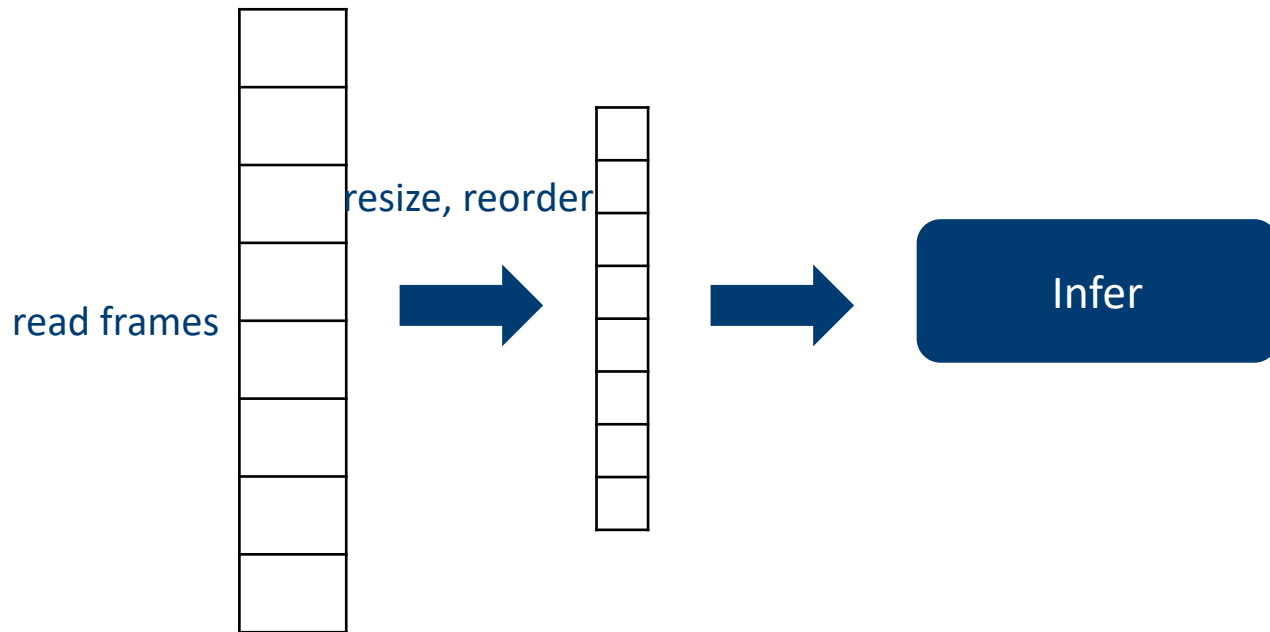
Tracking

Many ways to track:

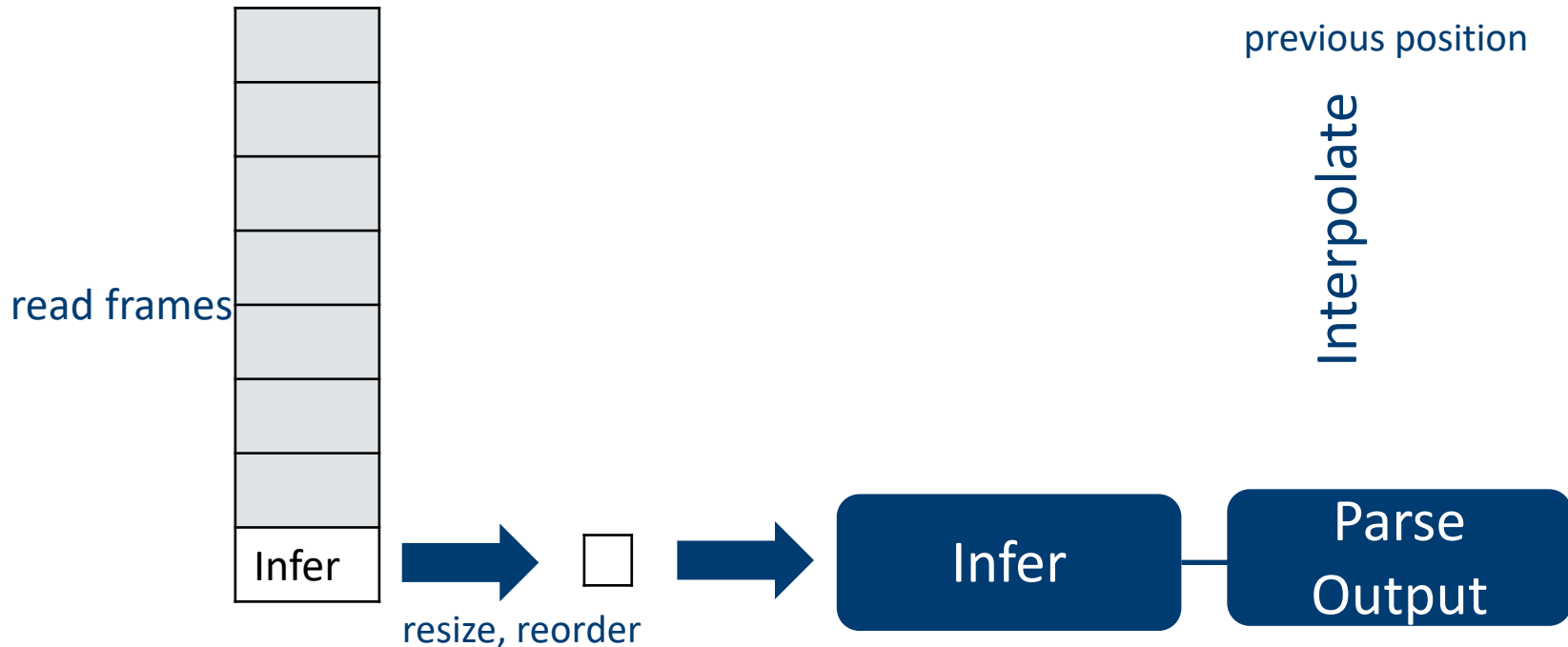
- Optical flow: the pattern of apparent motion of objects, surfaces, and edges in a visual scene caused by the relative motion between an observer and a scene.
- Kalman filters
- Single/simplistic tracking (what we show here)
- Multiple object tracking: like above but find ways to identify each region so rectangles have a persistent identity over time

A Tracking Thought Experiment

Processing in batches already introduces latency



What If We Did Less Inference?



Determine If There Is Nothing to See

Inference is expensive to run each frame. It can save time to not run when there is nothing to identify.

- Check motion vectors
- Frame sizes
- bgsbmog
- SAD

These methods can be several orders of magnitude less expensive than inference. Use techniques to increase the total # of streams a system can watch.



Lab5 - Optimizing Computer Vision Applications

URL: <https://github.com/intel-iot-devkit/smart-video-workshop/blob/master/optimization-tools-and-techniques/README.md>

Objective: This tutorial shows some techniques to get better performance for computer vision applications with the Intel® Distribution of OpenVINO™ toolkit.

Estimated Complete Time: 40min

Use Intel® VTune™ Analyzer

Use Intel® VTune™ Analyzer



Lab6 - Intel® VTune™ Amplifier Tutorial

URL: https://github.com/intel-iot-devkit/smart-video-workshop/blob/master/optimization-tools-and-techniques/README_VTune.md

Objective: This tutorial will show how to run Intel® VTune™ Amplifier on an Intel® Distribution of OpenVINO™ toolkit Inference Engine application.

Estimated Complete Time: 30min

ADVANCED VIDEO ANALYTICS

SECURITY BARRIER DEMO

Video Analytics in Intel® Distribution of OpenVINO™ Toolkit

Topology	Type	Description
<u>license-plate-recognition-barrier-0001</u>	ocr	Chinese license plate recognition.
<u>vehicle-attributes-recognition-barrier-0010</u>	object_attributes	Vehicle attributes recognition with modified RESNET10* backbone.
<u>vehicle-license-plate-detection-barrier-0007</u>	detection	Multiclass (vehicle, license plates) detector based on RESNET10 plus SSD.

vehicle-attributes-recognition-barrier-0010

Use Case/High-Level Description

Vehicle attributes classification algorithm for a traffic analysis scenario.

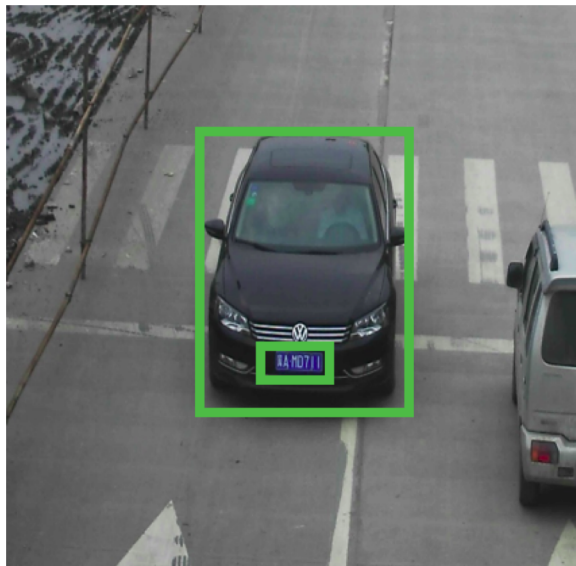


Type: regular
Color: black

vehicle-license-plate-detection-barrier-007

Use Case/High-Level Description

RESNET* 10 plus SSD-based vehicle and (Chinese) license plate detector for "Barrier" use case.



license-plate-recognition-barrier-0001

Use Case/High-Level Description

Small-footprint network trained E2E to recognize Chinese license plates in traffic scenarios.

Note: The license plates in the image are modified from the originals.



Lab7 - Advanced Video Analytics

URL: https://github.com/intel-iot-devkit/smart-video-workshop/blob/master/advanced-video-analytics/multiple_models.md

Objective: The tutorial shows some techniques for developing advanced video analytics applications.

Estimated Complete Time: 20min

